

Gamma Function

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1 Definition

Definition 1.1. *The Gamma Function is defined as:*

$$\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt$$

The first few evaluations of the Gamma function are:

$$\begin{aligned}\Gamma(1) &= \int_0^{\infty} e^{-t} dt = -e^{-t} \Big|_0^{\infty} = 0 - (-1) = 1 \\ \Gamma(2) &= \int_0^{\infty} e^{-t} t dt = -e^{-t} t \Big|_0^{\infty} + \int_0^{\infty} e^{-t} dt = 1 \\ \Gamma(3) &= \int_0^{\infty} e^{-t} t^2 dt = -e^{-t} t^2 \Big|_0^{\infty} + 2 \int_0^{\infty} e^{-t} t dt = 2 \\ \Gamma(4) &= \int_0^{\infty} e^{-t} t^3 dt = -e^{-t} t^3 \Big|_0^{\infty} + 3 \int_0^{\infty} e^{-t} t^2 dt = 6\end{aligned}\tag{1}$$

2 Relation to the Factorial Function

Expanding on the pattern we found by evaluating the first few integer values of the Gamma function through integration by parts we obtain:

Theorem 2.1. $\Gamma(z+1) = z\Gamma(z)$

Proof by induction

Assume true for k

$$\Gamma(k+1) = k\Gamma(k)$$

Proving for $k+1$

$$\begin{aligned}\Gamma(k+1) &= \int_0^{\infty} e^{-t} t^k dt \\ &= -e^{-t} t^k \Big|_0^{\infty} + k \int_0^{\infty} e^{-t} t^{k-1} dt \\ &= 0 - 0 + k\Gamma(k) \\ &= k\Gamma(k)\end{aligned}$$

2.1 Integer Values

Therefore since $\Gamma(1) = 1$, then for positive integer values

$$\Gamma(n) = (n-1)! \quad (2)$$

3 Equivalent Integral Forms

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt \quad (3)$$

Theorem 3.1. *Starting from Equation 1.1, we can use the substitution $t = -\ln(u)$ to obtain*

$$\Gamma(z) = \int_0^1 (-\ln u)^{z-1} du \quad (4)$$

Proof

$$\begin{aligned} t &= -\ln(u) \\ dt &= -\frac{du}{u} \\ \Gamma(z) &= \int_1^0 e^{\ln(u)} (-\ln(u))^{z-1} \frac{-du}{u} \\ \Gamma(z) &= - \int_1^0 u (-\ln(u))^{z-1} \frac{du}{u} \\ \Gamma(z) &= - \int_1^0 (-\ln(u))^{z-1} du \\ \Gamma(z) &= \int_0^1 (-\ln(u))^{z-1} du \end{aligned}$$

4 Finite Gamma Expression

Define the finite gamma function (find out its real name)

$$\Gamma_p(x) = \frac{p! p^x}{x(x+1)(x+2)\dots(x+p)} = \frac{p^x}{x(\frac{x}{1}+1)(\frac{x}{2}+1)\dots(\frac{x}{p}+1)} \quad x > 0 \quad (5)$$

Note that this function has the following properties:

$$\Gamma_p(1) = \frac{p! p}{1(1+1)(1+2)\dots(1+p)} = \frac{p! p}{p!(1+p)} = \frac{p}{1+p} \quad (6)$$

$$\begin{aligned} \Gamma_p(x+1) &= \frac{p! p p^x}{(x+1)(x+2)\dots(x+p)(x+p+1)} \\ &= \frac{x p! p p^x}{x(x+1)(x+2)\dots(x+p)(x+p+1)} \\ &= \frac{x p}{x+p+1} \Gamma_p(x) \end{aligned} \quad (7)$$

4.1 Taking the limit to obtain Gamma

Using Equation ?? and assuming $\Gamma_\infty(x)$ converges we get:

$$\lim_{p \rightarrow \infty} \Gamma_p(x+1) = \lim_{p \rightarrow \infty} \frac{xp}{x+p+1} \Gamma_p(x) = x \lim_{p \rightarrow \infty} \Gamma_p(x) \quad (8)$$

Therefore using the above properties we see that if $\Gamma_\infty(x)$ converges then it must be equal to $\Gamma(x)$.

5 Euler Mascheroni constant

The Euler Mascheroni constant is defined as:

$$\gamma = \lim_{p \rightarrow \infty} \sum_{i=1}^p 1/i - \ln(p) \approx 0.5772 \quad (9)$$

This constant can be interpreted as the difference between the sum of $1/x$ and the integral of $1/x$ from 1 to ∞ . It is unknown whether this number is transcendental or even irrational! One can show numerically that if it is rational then the denominator must be greater than 10^{242080} .

6 Weierstrass Formula

Theorem 6.1. *Let γ be the Euler Mascheroni constant as defined in section 5, and Γ be the Gamma function, then*

$$\frac{1}{\Gamma(x)} = x \cdot e^{x\gamma} \cdot \prod_{k=1}^{\infty} (1 + x/k) \cdot e^{-x/k} \quad (10)$$

Proof

Re-writing p^x

$$p^x = e^{x \ln p} = e^{x[\ln p - 1 - 1/2 - 1/3 - \dots - 1/p]} e^{x + x/2 + x/3 + \dots + x/p}$$

Plugging p^x into equation ??

$$\Gamma_p(x) = \frac{e^{x[\ln p - 1 - 1/2 - 1/3 - \dots - 1/p]} e^{x + x/2 + x/3 + \dots + x/p}}{x(\frac{x}{1} + 1)(\frac{x}{2} + 1) \dots (\frac{x}{p} + 1)}$$

$$\Gamma_p(x) = e^{x[\ln p - 1 - 1/2 - 1/3 - \dots - 1/p]} \cdot \frac{1}{x} \cdot \frac{e^x}{x+1} \cdot \frac{e^{x/2}}{x/2+1} \cdot \frac{e^{x/3}}{x/3+1} \cdot \dots \cdot \frac{e^{x/p}}{x/p+1}$$

$$\Gamma_p(x) = e^{x[\ln p - 1 - 1/2 - 1/3 - \dots - 1/p]} \cdot \frac{1}{x} \cdot \prod_{k=1}^p \frac{e^{x/k}}{1 + x/k}$$

Taking the limit as $p \rightarrow \infty$

$$\Gamma(x) = \lim_{p \rightarrow \infty} \Gamma_p(x) = \lim_{p \rightarrow \infty} e^{x[\ln p - 1 - 1/2 - 1/3 - \dots - 1/p]} \cdot \frac{1}{x} \cdot \prod_{k=1}^p \frac{e^{x/k}}{1 + x/k}$$

$$\Gamma(x) = e^{-x\gamma} \cdot \frac{1}{x} \cdot \prod_{k=1}^{\infty} \frac{e^{x/k}}{1 + x/k}$$

Equivalently

$$\frac{1}{\Gamma(x)} = x \cdot e^{x\gamma} \cdot \prod_{k=1}^{\infty} (1 + x/k) \cdot e^{-x/k}$$

6.1 Symmetry around 1/2 relation

Using 6 we derive the following functional form.

Theorem 6.2.

$$\boxed{\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(1-x)} = \frac{\sin(\pi x)}{\pi}} \quad (11)$$

Proof

Starting from 6 and considering $\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(-x)}$ we obtain

$$\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(-x)} = x \cdot e^{x\gamma} \cdot \prod_{k=1}^{\infty} (1 + x/k) \cdot e^{-x/k} \cdot -x \cdot e^{-x\gamma} \cdot \prod_{k=1}^{\infty} (1 - x/k) \cdot e^{x/k}$$

$$\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(-x)} = -x^2 \cdot \prod_{k=1}^{\infty} (1 - x^2/k^2)$$

Using the functional form of Γ , we get $\Gamma(1-x) = \Gamma(-x+1) = -x\Gamma(-x)$

$$\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(1-x)} = x \cdot \prod_{k=1}^{\infty} (1 - x^2/k^2)$$

Using the product form of $\frac{\sin(\pi x)}{\pi x} = \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2}\right)$, See Euler's Sine Product Form document

$$\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(1-x)} = \frac{\sin(\pi x)}{\pi}$$

6.2 Special Examples

Plugging in various values for x we obtain certain values and relations for $\Gamma(x)$

$$x = \frac{1}{2}$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\frac{\pi}{\sin(\frac{\pi}{2})}} = \sqrt{\pi} \quad (12)$$

$$x = \frac{1}{3}$$

$$\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{2}{3}\right) = \frac{\pi}{\sin(\frac{\pi}{3})} = \frac{2\pi}{\sqrt{3}} \quad (13)$$

$$x = \frac{1}{4}$$

$$\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = \frac{\pi}{\sin(\frac{\pi}{4})} = \sqrt{2}\pi \quad (14)$$

Theorem 6.3. $x = -\frac{1}{2}$ with equation ??

$$\Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi} \quad (15)$$

Proof

$$\Gamma\left(-\frac{1}{2}\right) \cdot \Gamma\left(1 - \left(-\frac{1}{2}\right)\right) = \frac{\pi}{\sin\left(-\frac{\pi}{2}\right)}$$

$$\Gamma\left(-\frac{1}{2}\right) = -\frac{\pi}{\Gamma\left(\frac{3}{2}\right)}$$

$$\Gamma\left(-\frac{1}{2}\right) = -\frac{\pi}{\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}$$

$$\Gamma\left(-\frac{1}{2}\right) = -\frac{\pi}{\frac{1}{2}\sqrt{\pi}}$$

$$\Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}$$

Note that equation ?? implies that

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty \frac{1}{e^t \sqrt{t}} dt = \sqrt{\pi} \quad (16)$$

Since $\Gamma(x) = (x-1)!$ for integer values of x , this is where the idea of $(-\frac{1}{2})! = \sqrt{\pi}$ comes from.

7 Other relations

Using equation 1.1 we derive the following.

Theorem 7.1. *This is the more general multiplication equation referred to as the multiplication formula or Gauss's multiplication formula*

$$\Gamma(x) \cdot \Gamma\left(x + \frac{1}{n}\right) \cdot \Gamma\left(x + \frac{2}{n}\right) \cdots \Gamma\left(x + \frac{n-1}{n}\right) = (2\pi)^{\frac{n-1}{2}} n^{1/2-nx} \Gamma(nx) \quad (17)$$

Proof

TODO

Plugging in $x = 0$ into Equation ?? we get

$$\Gamma\left(\frac{1}{n}\right) \cdot \Gamma\left(\frac{2}{n}\right) \cdots \Gamma\left(\frac{n-1}{n}\right) = \frac{(2\pi)^{\frac{n-1}{2}}}{\sqrt{n}} \quad (18)$$

Theorem 7.2. *There is a multiplicative relation to the Zeta function. Refer to the Zeta function document for the definition and properties of the Zeta function.*

$$\Gamma(x)\zeta(x) = \int_0^\infty \frac{u^{x-1}}{e^u - 1} du \quad (19)$$

Proof

See the Zeta function document.

8 Digamma function

8.1 Definition

Define the Digamma function

$$\Psi(x) = \frac{\partial}{\partial x} \ln(\Gamma(x)) = \frac{\Gamma'(x)}{\Gamma(x)} \quad (20)$$

8.2 Series Representation

Theorem 8.1. *Let γ be the Euler Mascheroni constant as defined in section 5 then,*

$$\Psi(x) = -\gamma - \frac{1}{x} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{x+k} \right) \quad (21)$$

Proof

$$\Psi(x) = \frac{\partial}{\partial x} \ln(\Gamma(x))$$

Using equation ??

$$\Psi(x) = -\frac{\partial}{\partial x} \ln(x \cdot e^{x\gamma} \cdot \prod_{k=1}^{\infty} (1 + x/k) \cdot e^{-x/k})$$

$$\Psi(x) = -\frac{\partial}{\partial x} \left[\ln(x) + x\gamma + \sum_{k=1}^{\infty} \ln(1 + x/k) - \frac{x}{k} \right]$$

$$\Psi(x) = -\frac{1}{x} - \gamma - \sum_{k=1}^{\infty} \frac{1}{k} \cdot \frac{1}{1 + x/k} - \frac{1}{k}$$

$$\Psi(x) = -\frac{1}{x} - \gamma + \sum_{k=1}^{\infty} \frac{1}{k} - \frac{1}{x+k}$$

8.3 Derivatives

Theorem 8.2.

$$\boxed{\Psi'(x) = \sum_{k=1}^{\infty} \frac{1}{(k+x-1)^2}} \quad (22)$$

Proof

Using equation ??

$$\Psi(x) = -\gamma - \frac{1}{x} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{x+k} \right)$$

$$\Psi'(x) = \frac{1}{x^2} + \sum_{k=1}^{\infty} \frac{1}{(x+k)^2}$$

Including the first term in the sum

$$\Psi'(x) = \sum_{k=1}^{\infty} \frac{1}{(x+k-1)^2}$$

Theorem 8.3.

$$\boxed{\Psi^n(x) = \sum_{k=1}^{\infty} \frac{(-1)^{n+1} n!}{(k+x-1)^{n+1}}} \quad (23)$$

Proof

Take repeated derivatives of equation ??

8.4 Special values

Theorem 8.4.

$$\Psi^n(x+1) = \Psi^n(x) + \frac{(-1)^n n!}{x^{n+1}} \quad (24)$$

Proof

$$\Psi^n(x+1) = \sum_{k=1}^{\infty} \frac{(-1)^{n+1} n!}{(k+x)^{n+1}}$$

$$\Psi^n(x+1) = \frac{(-1)^n n!}{x^{n+1}} + \sum_{k=2}^{\infty} \frac{(-1)^{n+1} n!}{(k+x)^{n+1}}$$

$$\Psi^n(x+1) = \frac{(-1)^n n!}{x^{n+1}} + \sum_{k=1}^{\infty} \frac{(-1)^{n+1} n!}{(k+x-1)^{n+1}}$$

$$\Psi^n(x+1) = \frac{(-1)^n n!}{x^{n+1}} + \Psi^n(x)$$

Theorem 8.5. *Reflection formula*

$$\Psi(1-x) = \Psi(x) + \pi \cot(\pi x) \quad (25)$$

Proof

$$\Psi(1-x) = -\gamma - \frac{1}{1-x} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1-x} \right)$$

Adding and subtracting $\Psi(x)$

$$\begin{aligned} \Psi(1-x) &= -\gamma - \frac{1}{1-x} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1-x} \right) + \Psi(x) - \Psi(x) \\ \Psi(1-x) &= -\gamma - \frac{1}{1-x} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1-x} \right) + \Psi(x) - \left(-\gamma - \frac{1}{x} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{x+k} \right) \right) \end{aligned}$$

Simplifying

$$\begin{aligned} \Psi(1-x) &= \Psi(x) + \frac{1}{x} - \frac{1}{1-x} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1-x} - \frac{1}{k} + \frac{1}{x+k} \right) \\ \Psi(1-x) &= \Psi(x) + \frac{1}{x} - \frac{1}{1-x} + \sum_{k=1}^{\infty} \left(\frac{1}{x+k} - \frac{1}{k+1-x} \right) \\ \Psi(1-x) &= \Psi(x) + \sum_{k=0}^{\infty} \left(\frac{1}{x+k} + \frac{1}{x-k-1} \right) \\ \Psi(1-x) &= \Psi(x) + \sum_{k=-\infty}^{\infty} \left(\frac{1}{x+k} \right) \end{aligned}$$

$$\text{Using the series }^1 \text{ for } \pi \cot(\pi x) = \sum_{k=-\infty}^{\infty} \frac{1}{x+k}$$

$$\Psi(1-x) = \Psi(x) + \pi \cot(\pi x)$$

Theorem 8.6.

$$\Psi(2x) = \frac{1}{2} \cdot \Psi(x) + \frac{1}{2} \cdot \Psi\left(x + \frac{1}{2}\right) + \ln(2) \quad (26)$$

Proof
TODO

$$\Psi(1) = \Gamma'(1) = -\gamma \quad (27)$$

$$\Psi^{(n)}(1) = (-1)^{n+1} n! \zeta(n+1) \quad (28)$$

Where $\zeta(z)$ is the Riemann Zeta function defined as $\zeta(z) = \sum_{k=1}^{\infty} \frac{1}{k^z}$. CONFIRM THIS?

$$\Psi(n) = \frac{\Gamma'(n)}{\Gamma(n)} = -\gamma + \sum_{k=1}^{n-1} \frac{1}{k} = -\gamma + H_{n-1} \quad (29)$$

Where H_n is the sum of the n first harmonic numbers. Note that this implies that the derivative of the Γ function is diverges to ∞ .

¹See the Euler's Sine Product formula document.

9 Using The Gamma Function for Euler Integrals

Theorem 9.1. Let $p = x + yi$ and $\alpha = \tan^{-1} \left(\frac{y}{x} \right)$ then

$$\boxed{\frac{\Gamma(s)}{n|p|^s} \sin(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} \sin(yu^n) du} \quad (30)$$

$$\boxed{\frac{\Gamma(s)}{n|p|^s} \cos(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} \cos(yu^n) du} \quad (31)$$

Proof

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt$$

Using substitution $t = p \cdot u^n$ hence $dt = p \cdot n \cdot u^{n-1} \cdot du$

$$\Gamma(s) = \int_0^\infty p^{n-1} u^{ns-n} e^{-pu^n} n p u^{n-1} du = n \int_0^\infty p^s u^{ns-1} e^{-pu^n} du$$

$$\frac{\Gamma(s)}{n p^s} = \int_0^\infty u^{ns-1} e^{-pu^n} du$$

Assuming p is a complex number with real part x , imaginary part y

Taking the conjugate of both sides, we obtain a second equation.

$$\frac{\Gamma(s)}{n \bar{p}^s} = \int_0^\infty u^{ns-1} e^{-\bar{p}u^n} du$$

Note that $p = x + yi = |p|e^{i\alpha}$ and $\bar{p} = x - yi = |p|e^{-i\alpha}$ where $\alpha = \tan^{-1} \left(\frac{y}{x} \right)$

Adding and subtracting both equations we get

$$\begin{aligned} \frac{\Gamma(s)}{n} \left[\frac{1}{\bar{p}^s} \pm \frac{1}{p^s} \right] &= \int_0^\infty u^{ns-1} \left[e^{-\bar{p}u^n} \pm e^{-pu^n} \right] du \\ \frac{\Gamma(s)}{n|p|^s} \left[\frac{1}{e^{-i\alpha s}} \pm \frac{1}{e^{i\alpha s}} \right] &= \int_0^\infty u^{ns-1} \left[e^{-xu^n + yu^ni} \pm e^{-xu^n - yu^ni} \right] du \\ \frac{\Gamma(s)}{n|p|^s} \left[e^{i\alpha s} \pm e^{-i\alpha s} \right] &= \int_0^\infty u^{ns-1} e^{-xu^n} \left[e^{yu^ni} \pm e^{-yu^ni} \right] du \end{aligned}$$

With addition we get

$$\begin{aligned} \frac{\Gamma(s)}{n|p|^s} \left[2\cos(\alpha s) \right] &= \int_0^\infty u^{ns-1} e^{-xu^n} \left[2\cos(yu^n) \right] du \\ \frac{\Gamma(s)}{n|p|^s} \cos(\alpha s) &= \int_0^\infty u^{ns-1} e^{-xu^n} \cos(yu^n) du \end{aligned}$$

With subtraction we get

$$\begin{aligned} \frac{\Gamma(s)}{n|p|^s} \left[2i\sin(\alpha s) \right] &= \int_0^\infty u^{ns-1} e^{-xu^n} \left[2i\sin(yu^n) \right] du \\ \frac{\Gamma(s)}{n|p|^s} \sin(\alpha s) &= \int_0^\infty u^{ns-1} e^{-xu^n} \sin(yu^n) du \end{aligned}$$

9.1 Examples

9.1.1 Sinc Integral $\int \frac{\sin(x)}{x}$

Theorem 9.2.

$$\boxed{\int_0^\infty \frac{\sin(x)}{x} = \frac{\pi}{2}} \quad (32)$$

Proof

Starting from equation ??

$$\frac{\Gamma(s)}{n|p|^s} \sin(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} \sin(yu^n) du$$

setting $y = 1, n = 1, x = 0$ and taking the limit as $s \rightarrow 0$ we obtain

$$|p| = 1, \alpha = \tan^{-1}(1/0) = \pi/2$$

$$\lim_{s \rightarrow 0} \Gamma(s) \sin(s \frac{\pi}{2}) = \lim_{s \rightarrow 0} \int_0^\infty u^{s-1} \sin(u) du = \int_0^\infty \frac{\sin(u)}{u} du$$

To evaluate the RHS limit we use equation ?? $\Gamma(s) = \frac{\pi}{\Gamma(1-s) \sin(\pi s)}$

$$\lim_{s \rightarrow 0} \Gamma(s) \sin(s \frac{\pi}{2}) = \lim_{s \rightarrow 0} \frac{\pi}{\Gamma(1-s) \sin(\pi s)} \sin(s \frac{\pi}{2})$$

We can split this limit into the product of 3 limits since all of these limits exist

$$\lim_{s \rightarrow 0} \Gamma(s) \sin(s \frac{\pi}{2}) = \lim_{s \rightarrow 0} \frac{\frac{\pi}{2}}{\Gamma(1-s)} \cdot \lim_{s \rightarrow 0} \frac{\pi}{\sin(\pi s)} \cdot \lim_{s \rightarrow 0} \frac{\sin(s \frac{\pi}{2})}{\frac{\pi}{2}}$$

$$\lim_{s \rightarrow 0} \Gamma(s) \sin(s \frac{\pi}{2}) = \frac{\pi}{2}$$

$$\int_0^\infty \frac{\sin(u)}{u} du = \frac{\pi}{2}$$

since $\frac{\sin(u)}{u}$ is an even function

$$\int_{-\infty}^\infty \frac{\sin(u)}{u} du = \pi$$

9.1.2 $\int \sin(x^2)$

Theorem 9.3.

$$\boxed{\int_0^\infty \sin(x^2) = \frac{\sqrt{2\pi}}{4}} \quad (33)$$

Proof

Starting from equation ??

$$\frac{\Gamma(s)}{n|p|^s} \sin(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} \sin(yu^n) du$$

setting $y = 1, n = 2, x = 0$ and $ns - 1 = 0$ we obtain

$$|p| = 1, \alpha = \tan^{-1}(1/0) = \pi/2, s = 1/2$$

$$\frac{\Gamma(\frac{1}{2}) \sin(\frac{\pi}{4})}{2} = \int_0^\infty \sin(u^2) du$$

$$\text{using equation ??: } \Gamma(\frac{1}{2}) = \sqrt{\pi}$$

$$\int_0^\infty \sin(u^2) du = \frac{\sqrt{\pi} \frac{1}{\sqrt{2}}}{2} = \frac{\sqrt{2\pi}}{4}$$

9.1.3 $\int \cos(x^2)$

Theorem 9.4.

$$\boxed{\int_0^\infty \cos(x^2) = \frac{\sqrt{2\pi}}{4}} \quad (34)$$

Proof

Starting from equation ??

$$\frac{\Gamma(s)}{n|p|^s} \cos(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} \cos(yu^n) du$$

setting $y = 1, n = 2, x = 0$ and $ns - 1 = 0$ we obtain

$$|p| = 1, \alpha = \tan^{-1}(1/0) = \pi/2, s = 1/2$$

$$\frac{\Gamma(\frac{1}{2}) \cos(\frac{\pi}{4})}{2} = \int_0^\infty \cos(u^2) du$$

$$\text{using equation ??: } \Gamma(\frac{1}{2}) = \sqrt{\pi}$$

$$\int_0^\infty \cos(u^2) du = \frac{\sqrt{\pi} \frac{1}{\sqrt{2}}}{2} = \frac{\sqrt{2\pi}}{4}$$

9.1.4 $\int \frac{\sin(x^2)}{x^2}$

Theorem 9.5.

$$\boxed{\int_0^\infty \frac{\sin(x^2)}{x^2} = \sqrt{\frac{\pi}{2}}} \quad (35)$$

Proof

Starting from equation ??

$$\frac{\Gamma(s)}{n|p|^s} \sin(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} \sin(yu^n) du$$

setting $y = 1, n = 2, x = 0$ and $ns - 1 = -2$ we obtain

$$|p| = 1, \alpha = \tan^{-1}(1/0) = \pi/2, s = -1/2$$

$$\frac{\Gamma(-\frac{1}{2}) \sin(-\frac{\pi}{4})}{2} = \int_0^\infty \frac{\sin(u^2)}{u^2} du$$

$$\text{using equation ??: } \Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$$

$$\int_0^\infty \frac{\sin(u^2)}{u^2} du = -\frac{\Gamma(-\frac{1}{2})}{2\sqrt{2}} = \sqrt{\frac{\pi}{2}}$$

9.1.5 $\int \frac{\sin(x^n)}{x^n}$

Theorem 9.6.

$$\boxed{\int_0^\infty \frac{\sin(x^n)}{x^n} = \frac{\Gamma(\frac{1}{n})}{n-1} \cdot \cos(\frac{\pi}{2n}) \quad , n > 1} \quad (36)$$

Proof

Starting from equation ??

$$\frac{\Gamma(s)}{n|p|^s} \sin(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} \sin(yu^n) du$$

setting $y = 1, x = 0$ and $ns - 1 = -n$ we obtain

$$|p| = 1, \alpha = \tan^{-1}(1/0) = \pi/2, s = 1/n - 1$$

$$\frac{\Gamma(\frac{1}{n} - 1) \sin(\frac{\pi}{2n} - \frac{\pi}{2})}{n} = \int_0^\infty \frac{\sin(u^n)}{u^n} du$$

using equation 2.1 and assuming $n \neq 1$: $\Gamma(k+1) = k\Gamma(k)$

$$\int_0^\infty \frac{\sin(u^n)}{u^n} du = \frac{\Gamma(\frac{1}{n})}{n(\frac{1}{n} - 1)} \sin(\frac{\pi}{2n} - \frac{\pi}{2})$$

$$\int_0^\infty \frac{\sin(u^n)}{u^n} du = \frac{\Gamma(\frac{1}{n})}{1-n} \sin(\frac{\pi}{2n} - \frac{\pi}{2})$$

$$\text{using } \sin\left(x - \frac{\pi}{2}\right) = -\cos(x)$$

$$\int_0^\infty \frac{\sin(u^n)}{u^n} du = \frac{\Gamma(\frac{1}{n})}{n-1} \cdot \cos(\frac{\pi}{2n})$$

10 Relation to the Beta Function

10.1 Definition

The Beta Function is defined as:

Definition 10.1.

$$\beta(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt$$

10.2 Symmetry

Theorem 10.1.

$$\beta(x, y) = \beta(y, x) \quad (37)$$

Proof

Starting from the definition [10.1](#)

$$\beta(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt$$

setting $u = 1 - t$, hence $du = -dt$ we obtain

$$\beta(x, y) = \int_1^0 (1-u)^{x-1} u^{y-1} - du$$

Switching the bounds of integration we get

$$\beta(x, y) = \int_0^1 (1-u)^{x-1} u^{y-1} du = \beta(y, x)$$

10.3 Relation to Gamma

Theorem 10.2.

$$\boxed{\beta(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}} \quad (38)$$

Proof

Starting from the definition [10.1](#)

$$\beta(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt$$

setting $t = \frac{1}{1+u}$, hence $dt = -\frac{1}{(1+u)^2} du$ we obtain

$$\beta(x, y) = \int_{\infty}^0 \left(\frac{1}{1+u}\right)^{x-1} \left(1 - \frac{1}{1+u}\right)^{y-1} \frac{1}{(1+u)^2} (-du)$$

Switching the bounds of integration and simplifying we get

$$\beta(x, y) = \int_0^{\infty} \left(\frac{1}{1+u}\right)^{x-1} \left(\frac{u}{1+u}\right)^{y-1} \frac{1}{(1+u)^2} du$$

collecting powers of $(1+u)$

$$\beta(x, y) = \int_0^{\infty} \frac{u^{y-1}}{(1+u)^{x+y}} du$$

Using the Gamma definition [1.1](#) and applying a change of variables $\alpha w = t$, hence $dt = \alpha dw$

$$\Gamma(z) = \alpha^z \int_0^{\infty} e^{-\alpha w} w^{z-1} dw$$

setting $z = x + y$ and $\alpha = 1 + u$, we obtain

$$\Gamma(x + y) = (1 + u)^{x+y} \int_0^{\infty} e^{-(1+u)w} w^{x+y-1} dw$$

Combining with the Beta function integral we obtained

$$\beta(x, y) \Gamma(x + y) = \int_0^{\infty} \Gamma(x + y) \frac{u^{y-1}}{(1+u)^{x+y}} du$$

$$\beta(x, y) \Gamma(x + y) = \int_0^{\infty} \left((1+u)^{x+y} \int_0^{\infty} e^{-(1+u)w} w^{x+y-1} dw \right) \frac{u^{y-1}}{(1+u)^{x+y}} du$$

Simplifying

$$\beta(x, y) \Gamma(x + y) = \int_0^{\infty} \int_0^{\infty} e^{-(1+u)w} w^{x+y-1} u^{y-1} dw du$$

Rearranging

$$\beta(x, y) \Gamma(x + y) = \int_0^{\infty} \left(\int_0^{\infty} e^{-uw} u^{y-1} du \right) e^{-w} w^{x+y-1} dw$$

$$\text{Using } \Gamma(z) = \alpha^z \int_0^{\infty} e^{-\alpha u} u^{z-1} du$$

$$\beta(x, y) \Gamma(x + y) = \int_0^{\infty} \frac{\Gamma(y)}{w^y} e^{-w} w^{x+y-1} dw$$

Simplifying and moving $\Gamma(y)$ outside the integral

$$\beta(x, y) \Gamma(x + y) = \Gamma(y) \int_0^{\infty} e^{-w} w^{x-1} dw$$

Using the definition of the $\Gamma(z)$

$$\beta(x, y) \Gamma(x + y) = \Gamma(y) \Gamma(x)$$

$$\beta(x, y) = \frac{\Gamma(y) \Gamma(x)}{\Gamma(x + y)}$$

10.4 Relation to integral of powers of $\sin(x)$ and $\cos(x)$

Theorem 10.3.

$$\boxed{\int_0^{\pi/2} \sin^n(t) \cos^m(t) dt = \frac{1}{2} \beta\left(\frac{n+1}{2}, \frac{m+1}{2}\right) = \frac{\Gamma(\frac{n+1}{2}) \Gamma(\frac{m+1}{2})}{\Gamma(\frac{n+m}{2} + 1)}} \quad (39)$$

Proof

$$A(m, n) = \int_0^{\pi/2} \sin^n(t) \cos^m(t) dt$$

setting $\sin^2(t) = u$, hence $2 \sin(t) \cos(t) dt = du$ we obtain

note $\cos^2(t) = 1 - u$, and $\sin^2(0) = 0$ and $\sin^2\left(\frac{\pi}{2}\right) = 1$

$$A(m, n) = \int_0^1 u^{\frac{n}{2}} (1-u)^{\frac{m}{2}} \frac{du}{2\sqrt{u}\sqrt{1-u}}$$

Simplifying the powers we get

$$A(m, n) = \frac{1}{2} \int_0^1 u^{\frac{n}{2}-\frac{1}{2}} (1-u)^{\frac{m}{2}-\frac{1}{2}} du$$

Writing it in the Beta function form we get

$$A(m, n) = \frac{1}{2} \int_0^1 u^{\frac{n+1}{2}-1} (1-u)^{\frac{m+1}{2}-1} du = \frac{1}{2} \beta\left(\frac{n+1}{2}, \frac{m+1}{2}\right)$$

10.5 Proof of the Legendre duplication formula

Theorem 10.4.

$$\boxed{\Gamma(x) \cdot \Gamma\left(x + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2x-1}} \Gamma(2x)} \quad (40)$$

Proof

Using the relation to the Beta function, and the definition of the Beta function

$$\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \int_0^1 t^{x-1}(1-t)^{y-1} dt$$

let $x = y = s$

$$\frac{\Gamma(s)\Gamma(s)}{\Gamma(2s)} = \int_0^1 t^{s-1}(1-t)^{s-1} dt$$

$$\text{let } t = \frac{1+x}{2}, \text{ hence } dt = \frac{dx}{2}$$

note when $t = 0$ then $x = -1$, and when $t = 1$ then $x = 1$

$$\frac{\Gamma(s)\Gamma(s)}{\Gamma(2s)} = \int_{-1}^1 \left(\frac{1+x}{2}\right)^{s-1} \left(\frac{1-x}{2}\right)^{s-1} \frac{dx}{2}$$

Simplifying the powers of 2 and rearranging

$$2^{2s-1}\Gamma(s)\Gamma(s) = \Gamma(2s) \int_{-1}^1 (1+x)^{s-1}(1-x)^{s-1} dx$$

$$2^{2s-1}\Gamma(s)\Gamma(s) = \Gamma(2s) \int_{-1}^1 (1-x^2)^{s-1} dx$$

Notice that the integral is even around 0, hence

$$2^{2s-1}\Gamma(s)\Gamma(s) = 2\Gamma(2s) \int_0^1 (1-x^2)^{s-1} dx$$

To evaluate the integral, notice that after a substitution of $u = x^2$ the integral is just $\beta(1/2, s)$

$$\beta(1/2, s) = \int_0^1 t^{-\frac{1}{2}}(1-t)^{s-1} dt$$

Let $t = x^2$, hence $dt = 2x dx$

$$\beta(1/2, s) = \int_0^1 \frac{1}{x} (1-x^2)^{s-1} 2x dx = 2 \int_0^1 (1-x^2)^{s-1} dx$$

Hence

$$2^{2s-1}\Gamma(s)\Gamma(s) = \Gamma(2s)\beta(1/2, s)$$

Using theorem [10.2](#)

$$2^{2s-1}\Gamma(s)\Gamma(s) = \Gamma(2s) \frac{\Gamma(1/2)\Gamma(s)}{\Gamma(s+1/2)}$$

Rearranging and using equation ??

$$\Gamma(s)\Gamma(s+1/2) = \Gamma(2s) \frac{\sqrt{\pi}}{2^{2s-1}}$$

11 Approximations

11.1 Stirling's Approximation

Theorem 11.1. *The basic approximation of the factorial function for large n is*

$$\boxed{n! \approx \left(\frac{n}{e}\right)^n \sqrt{2\pi n}} \quad (41)$$

Proof

Starting from the definition of the Gamma function [1.1](#)

$$\Gamma(n+1) = n! = \int_0^\infty e^{-t} t^n dt$$

the idea is to take the log of the integrand and then slightly perturb it

$$\ln(e^{-t} t^n) = n \ln(t) - t$$

$$\text{let } t = n + \epsilon$$

$$\ln(e^{-t} t^n) = n \ln(n + \epsilon) - (n + \epsilon) = n \ln(n) + n \ln\left(1 + \frac{\epsilon}{n}\right) - (n + \epsilon)$$

$$\text{for large } n, \frac{\epsilon}{n} \ll 1$$

Using the Taylor-Expansion of $\ln(1+x)$

$$\ln(e^{-t} t^n) = n \left(\ln(n) + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \left(\frac{\epsilon}{n}\right)^k \right) - (n + \epsilon)$$

$$\ln(e^{-t} t^n) = n \ln(n) - n + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \left(\frac{\epsilon^k}{n^{k-1}}\right) - \epsilon$$

$$\ln(e^{-t} t^n) = n \ln(n) - n - \frac{\epsilon^2}{2n} + \frac{\epsilon^3}{3n^2} - \frac{\epsilon^4}{4n^3} + \dots$$

$$\ln(e^{-t} t^n) \approx n \ln(n) - n - \frac{\epsilon^2}{2n}$$

$$e^{-t} t^n \approx \frac{n^n}{e^n} e^{-\frac{\epsilon^2}{2n}}$$

$$n! \approx \int_{-n}^{\infty} \frac{n^n}{e^n} e^{-\frac{\epsilon^2}{2n}} d\epsilon$$

$$n! \approx \frac{n^n}{e^n} \int_{-n}^{\infty} e^{-\frac{\epsilon^2}{2n}} d\epsilon$$

Assuming n is large enough

$$n! \approx \frac{n^n}{e^n} \int_{-\infty}^{\infty} e^{-\frac{\epsilon^2}{2n}} d\epsilon$$

$$\text{Using } \int_{-\infty}^{\infty} e^{-px^2} dx = \sqrt{\frac{\pi}{p}}$$

$$n! \approx \frac{n^n}{e^n} \sqrt{2n\pi}$$

11.1.1 Limit as n approaches infinity

The approximation in the this result can be stated more correctly as follows

$$\lim_{n \rightarrow \infty} \frac{n!}{\left(\frac{n}{e}\right)^n \sqrt{2\pi n}} = 1 \quad (42)$$

We can state this equivalently by taking logarithms of both sides. We will use this form to help derive the full Stirling approximation.

$$\lim_{n \rightarrow \infty} \ln(n!) - n \ln(n) + n - \frac{\ln(n)}{2} = \ln(\sqrt{2\pi}) \quad (43)$$

11.2 Stirling's Full Approximation

Theorem 11.2. *Expanding on Stirling's approximation to include as many terms as needed we need to use the Euler-Maclaurin formula².*

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \cdot \exp\left(\sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} - \frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt\right) \quad (44)$$

Proof

Starting from the definition of the simple version of the Euler-Maclaurin formula

$$\begin{aligned} \sum_{j=0}^r g(j) - \int_{x=0}^r g(x) dx &= \frac{g(0) + g(r)}{2} + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{(2k)!} \left(g^{(2k-1)}(r) - g^{(2k-1)}(0) \right) + R_{r,m} \\ R_{r,m} &= \frac{(-1)^{m+1}}{m!} \int_0^r B_m(\{t\}) g^{(m)}(t) dt \end{aligned}$$

Define $f(x) = g(x+1)$ and $n = r+1$, so that $f(n) = f(r+1) = g(r)$. we get

$$\begin{aligned} \sum_{j=1}^n f(j) - \int_{x=1}^n f(x) dx &= \frac{f(1) + f(n)}{2} + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{(2k)!} \left(f^{(2k-1)}(n) - f^{(2k-1)}(1) \right) + R_{n,m} \\ R_{n,m} &= \frac{(-1)^{m+1}}{m!} \int_1^{n-1} B_m(\{t\}) f^{(m)}(t) dt \end{aligned}$$

²See the Darboux Formula document

Let $f(x) = \ln(x)$, therefore

$$f^{(m)}(x) = \frac{(-1)^{m-1} \cdot (m-1)!}{(x)^m} \text{ or } f^{(2k-1)}(x) = \frac{(-1)^{2k-2} \cdot (2k-2)!}{(x)^{2k-1}}$$

Plugging in $f(x)$ and $f^{(m)}(x)$ into the Euler-Maclaurin formula we get

$$\begin{aligned} \sum_{j=1}^n \ln(j) &= \ln(n!) = \int_{x=1}^n \ln(x) dx + \frac{\ln(1) + \ln(n)}{2} \\ &+ \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{(2k)!} \left(\frac{(-1)^{2k-2} \cdot (2k-2)!}{n^{2k-1}} - \frac{(-1)^{2k-2} \cdot (2k-2)!}{1^{2k-1}} \right) + R_{n,m} \\ R_{n,m} &= \frac{(-1)^{m+1}}{m!} \int_1^{n-1} B_m(\{t\}) \frac{(-1)^{m-1} \cdot (m-1)!}{t^m} dt \end{aligned}$$

Evaluating the integral and simplifying

$$\begin{aligned} \ln(n!) &= (n \cdot \ln(n) - n) - (1 \cdot \ln(1) - 1) + \frac{\ln(n)}{2} + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k} \cdot (2k-2)!}{(2k)!} \left(\frac{1}{n^{2k-1}} - 1 \right) + R_{n,m} \\ R_{n,m} &= \frac{1}{m} \int_1^{n-1} \frac{B_m(\{t\})}{t^m} dt \end{aligned}$$

Further simplifying

$$\ln(n!) = n \cdot \ln(n) - n + 1 + \frac{\ln(n)}{2} + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \left(\frac{1}{n^{2k-1}} - 1 \right) + R_{n,m}$$

To eliminate some terms we can rearranging and take limits of both sides. This would allow us to use the simple Stirling Approximation to skip the evaluation of one of the sums.

$$\begin{aligned} &\lim_{n \rightarrow \infty} \ln(n!) - n \cdot \ln(n) + n - \frac{\ln(n)}{2} \\ &= \lim_{n \rightarrow \infty} 1 + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} - \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} + R_{n,m} \\ &= 1 - \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} + \lim_{n \rightarrow \infty} R_{n,m} \end{aligned}$$

Using equation ??

$$1 - \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} + \lim_{n \rightarrow \infty} R_{n,m} = \ln(\sqrt{2\pi})$$

Plugging this back in, we get:

$$\ln(n!) - n \cdot \ln(n) + n - \frac{\ln(n)}{2} = \ln(\sqrt{2\pi}) + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} + R_{n,m} - \lim_{n \rightarrow \infty} R_{n,m}$$

Using the integral form of $R_{n,m}$ we can then re-write the difference $D = R_{n,m} - \lim_{n \rightarrow \infty} R_{n,m}$

$$D = R_{n,m} - \lim_{n \rightarrow \infty} R_{n,m} = \frac{1}{m} \int_1^{n-1} \frac{B_m(\{t\})}{t^m} dt - \frac{1}{m} \int_1^{\infty} \frac{B_m(\{t\})}{t^m} dt = -\frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt$$

Now raising e to the value of both sides, we get

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \cdot \exp\left(\sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} - \frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt\right)$$

11.2.1 Approximation big-O notation

Theorem 11.3. *The integral in the exponent can be bound by a polynomial power with respect to n .*

$$\boxed{n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \cdot \exp\left(\sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} + O\left(\frac{1}{n^m}\right)\right)} \quad (45)$$

Proof

We need to show that $\frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt = O\left(\frac{1}{n^m}\right)$

$$\begin{aligned} D &= -\frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt \\ D \leq |D| &\leq \frac{1}{|m|} \int_{n-1}^{\infty} \frac{|B_m(\{t\})|}{t^m} dt \leq \frac{\max_{x \in [0,1]} |B_m(x)|}{|m|} \int_{n-1}^{\infty} \frac{1}{t^m} dt \\ &= \underbrace{\frac{\max_{x \in [0,1]} |B_m(x)|}{m \cdot (m+1)}}_{\neq f(n)} \underbrace{\frac{1}{(n-1)^m}}_{f(n)} = O\left(\frac{1}{n^m}\right) \end{aligned}$$

11.3 Plugging in values

Generating an approximation with $m = 7$ and using $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$ we obtain:

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \cdot \exp\left(\frac{1}{12n} - \frac{1}{360n^3} + \frac{1}{1260n^5} + O\left(\frac{1}{n^7}\right)\right) \quad (46)$$

Since the factorial function grows very fast, it's useful to consider the logarithm of the factorial function. This would result in the following approximation.

$$\ln(n!) = \frac{\ln(2\pi)}{2} + n \ln(n) - n + \frac{\ln(n)}{2} + \frac{1}{12n} - \frac{1}{360n^3} + \frac{1}{1260n^5} + O\left(\frac{1}{n^7}\right) \quad (47)$$